RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering — Cyber Security

CR23532

Cyber and Digital Forensics

CASE STUDY ASSIGNMENTS – II

Assignments 2(1) to 2(5) — unique incident scenarios with forensic analysis, flowcharts, evidence tables, timelines and reports

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Deliverables map

Assignment	Required deliverables	Where to find them
2(1)	Investigation flowchart + evidence table + case analysis + sample report structure	Fig 1.1 · Q3 table · Q1, Q5 · Q6 table
2(2)	System inventory table + evidence-linkage matrix + ILookIX workflow + conclusion with limitations	Q1 table · Q4 matrix · Q5 · Q7 + Conclusion
2(3)	Acquisition flowchart + chain-of-custody form + imaging procedure + evidence-integrity explanation	Fig 3.1 · Q8 form · Q3 table · Q4, Q5
2(4)	Evidence classification table + event timeline + VSC recovery workflow + validation checklist	Q2 table · Q6 table · Q5 · Q7 checklist
2(5)	Multi-source evidence map + correlation matrix + challenge analysis + final investigative summary	Fig 5.1 · Q6 matrix · Q7 table · Q8

Conventions used throughout

- Every scenario is unique and fictional. Names of people, organisations, devices, accounts, IP addresses, serial numbers and hash values were invented for this submission.
- Times are given in IST (UTC+05:30) unless marked "Z" or "UTC". Where clocks were wrong, both the raw and corrected values are explained.
- As required by the assignment note, tool procedures (IXImager, ILookIX and others) describe the intended forensic workflow; they are not claims that an operation was performed on real media.
- Each assignment answers every question in order (Q1, Q2, …), states its assumptions in an amber box at the start, and closes with a short reflection. Sources are cited as [n] and listed under References.

ASSIGNMENT – 2(1)

Suspected Unauthorized Access to a College Server

> **Unique Scenario: "Operation Night Shift"**. After-hours copying of end-semester question papers from the Mechanical Engineering exam-cell file server (MECH-FS01) at Kaveri Institute of Technology, Hosur, a fictional institution.
>
> Case No. KIT/DFC/2026/017 · Complaint received: Mon 14 Sep 2026, 09:20 IST · Examiner: M. M. Chandhan Sai, Digital Forensics Cell

Case Scenario

On the morning of Monday, 14 September 2026, Dr. R. Kalaivani, Exam Cell Coordinator for Mechanical Engineering, was going through the weekly audit report that the file server MECH-FS01 sends her every Monday at 08:00. Six entries stood out. They were read operations on the restricted folder `\\MECH-FS01\ExamCell\ESE-Nov2026\QP-Final\`, made on **Saturday, 12 September, between 22:49 and 22:55 IST**, by the account `KIT\mech.labadmin`. The campus is closed on weekends, and only she and the Head of Department are meant to open that folder. She lodged a written complaint with the Principal at 09:20.

The IT team traced the source address, 172.16.40.57, to **MECH-CADLAB-PC07**, a desktop in the CAD Laboratory (Room M-204). When the lab was opened at 11:05 the machine was switched off, and a **SanDisk Cruzer Blade 32 GB** pen drive was still sitting in its rear USB port. A look at group memberships explained how the account got in: `mech.labadmin`, which three lab staff share for installing CAD software, had been added to the `ExamCell-RW` group by mistake during a software rollout in August.

Why I chose this scenario. Question-paper leaks are a genuine worry for any college, and this case lets every phase of the forensic process be shown on systems a student knows well: a departmental server, a lab PC and a pen drive. It also carries a built-in complication, a shared account, which forces the analysis to keep "which account" and "which person" apart.

Item	Detail
Server	MECH-FS01, Windows Server 2022, SMB share `ExamCell`, advanced audit policy (File Share, File System) enabled
Files at risk	Six final question papers, e.g. `ME3491_Thermal_Engg_SetA.pdf`, `ME3491_Thermal_Engg_SetB.pdf`, `ME3592_Design_Machine_Elements_SetA.docx`
Account used	`KIT\mech.labadmin`, shared by three CAD-lab staff and wrongly placed in ExamCell-RW
Suspect workstation	MECH-CADLAB-PC07, Windows 11 Education 23H2, 256 GB NVMe SSD, domain-joined
Removable media	SanDisk Cruzer Blade 32 GB, FAT32, found inserted in PC07
Window of interest	Sat 12 Sep 2026, 18:00 to Sun 13 Sep 2026, 06:00 IST
Other sources	Domain controller logs, DHCP server log, RFID door-access log (Block M), corridor CCTV

> **Assumptions**
>
> - All names, devices, IP addresses and values are fictional and were created for this assignment.
> - Every device involved is college-owned and covered by the KIT IT Acceptable Use Policy, which permits inspection of institutional systems.
> - Server logs are kept for 90 days; the RFID and CCTV systems are run by the Estate Office.
> - Times are in IST (UTC+05:30) unless marked "Z" (UTC). Tool steps describe the intended workflow, and artefact values are illustrative.

Q1. Moving Through the Five Forensic Phases

The team works through the five phases in order, and each phase hands a defined output to the next [1], [4]. Keeping the phases distinct is what allows every conclusion in the final report to be traced back to a specific item of preserved evidence.

Phase	What the team does in this case	Output
Identification	Records the complaint and opens case KIT/DFC/2026/017 with the Principal's written authorisation. Fixes the scope (12–13 Sep; MECH-FS01, PC07, the pen drive) and lists every possible source of evidence, ranked by volatility. The server Security log can roll over within days, so it is first in line. IT staff are told not to log on to PC07 or plug the pen drive in anywhere, and Room M-204 is locked and photographed.	Case file, authorisation letter, prioritised list of evidence sources
Collection	Because exam work continues on the server, it cannot be shut down. The team performs a targeted live collection instead: it exports `Security.evtx`, the SMB audit logs and folder permissions, and hashes each export and the six original papers in place. DC and DHCP logs are exported next. PC07's SSD is removed and imaged through a hardware write blocker, and the pen drive through a USB write blocker. The Estate Office supplies hashed RFID and CCTV exports. Every item is sealed and labelled, and a chain-of-custody form is opened for it.	Forensic images, log exports, hash register, chain-of-custody forms
Examination	Image hashes are verified before anything else. The PC07 file system (`$MFT`, `$UsnJrnl`), registry hives (SYSTEM\USBSTOR, MountedDevices, the user's NTUSER.DAT), LNK files, Jump Lists, ShellBags, Prefetch and event logs are parsed. On the pen drive, the FAT32 directory is parsed, deleted entries are recovered and every file is hashed. Known operating-system files are filtered out with a known-good hash set.	Extracted artefacts, hash comparisons, reduced data set
Analysis	Server read events are matched against PC07 artefacts and the pen drive's contents, after all clocks are brought to IST. Three hypotheses are tested: H1, someone used PC07 in person; H2, PC07 was used remotely; H3, malware was responsible. Logon types, remote-access logs and running services decide which survive.	Correlated timeline, tested hypotheses, findings with stated confidence
Presentation	A forensic report is written with exhibits, a hash register and a timeline chart, and the Examination Committee is briefed in plain language. If the college refers the matter to the police, a Section 63 certificate under the Bharatiya Sakshya Adhiniyam, 2023 is attached.	Signed report, committee briefing, certificate

Q2. Forensic Investigation Flowchart

Figure 1.1 follows the investigation from the complaint through to the presentation of findings. Each colour band is one phase, and the diamonds mark points where the team must stop and record a decision instead of acting on assumption.

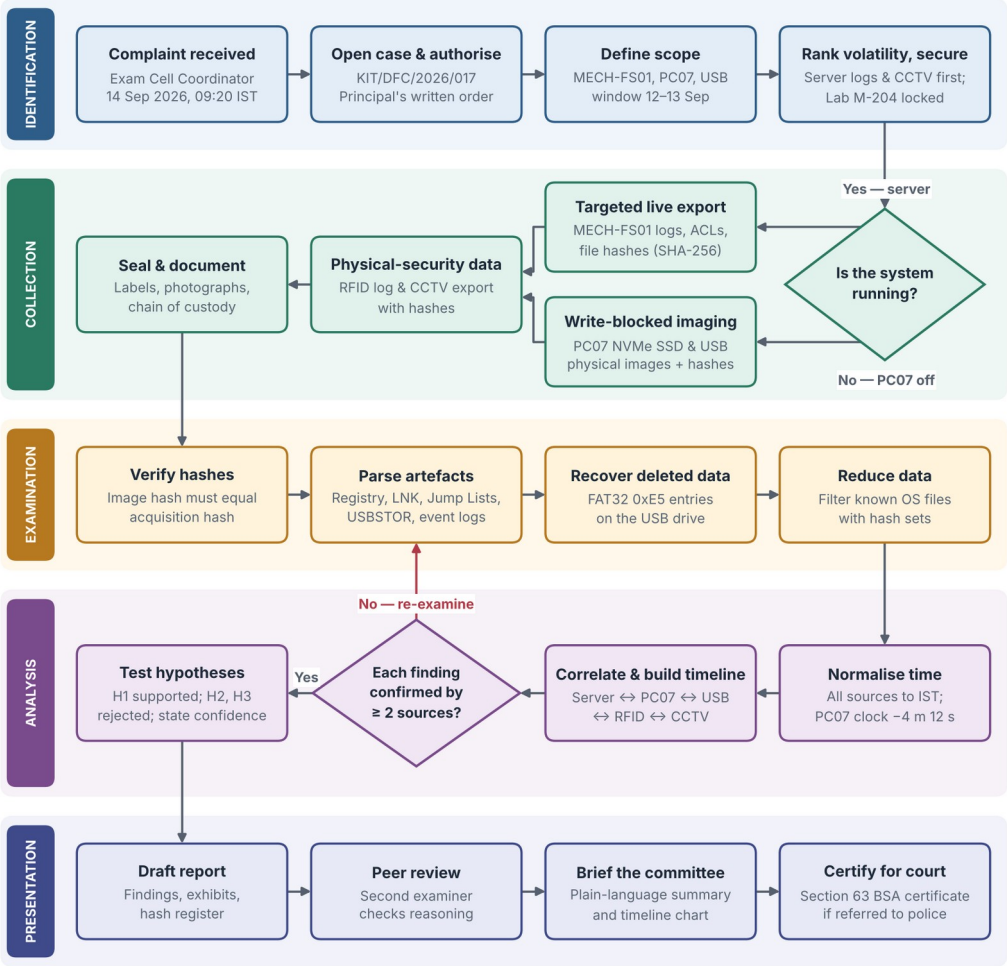


Figure 1.1: Forensic investigation flowchart for Case KIT/DFC/2026/017

Two decisions shape this particular case. The server is still needed for exam work, so it cannot be switched off; its logs are collected live and selectively, and the examiner notes that their own logon adds fresh entries to the Security log so those entries are not later mistaken for the suspect's. PC07, by contrast, was already off when found. With no memory left to capture, it goes straight to a write-blocked physical image.

Q3. Potential Digital Evidence and Its Classification

Ten items of potential evidence were identified. I have classified each by its investigative value on three levels: **primary** evidence shows the act itself, **corroborative** evidence independently confirms a primary item, and **contextual** evidence explains opportunity or circumstance without proving the act on its own.

ID	Evidence item	What it can show	Volatility	Collection method	Value
E1	Written complaint and Monday audit report	The allegation itself: which files, which account, which time window	Low	Signed original plus scanned PDF, hashed	Contextual (lead)
E2	MECH-FS01 Security log (Event IDs 4624, 5140, 5145, 4663)	Which account read which file, when, and from which IP address	**High**, log rolls over	Live export of `.evtx`, SHA-256	**Primary**
E3	Domain controller logs (4768/4769) and DHCP log	Links the `mech.labadmin` logon and IP 172.16.40.57 to PC07's MAC address	Medium	Export from DC and DHCP server, hashed	Corroborative
E4	PC07 NVMe SSD (forensic image)	Logon session, USB connection, LNK and Jump List entries for the papers, 7-Zip use	Low	Physical image via write blocker	**Primary**

ID	Evidence item	What it can show	Volatility	Collection method	Value
E5	SanDisk pen drive (forensic image)	Copies of the papers, the ZIP archive, deleted entries, FAT32 timestamps	Low	Physical image via USB write blocker	**Primary**
E6	Original question papers on MECH-FS01	Reference hashes proving the pen-drive copies are identical	Low	SHA-256 computed in place	**Primary** (reference)
E7	RFID door-access log, Block M	Whose card opened the building door, and when	Medium (30-day retention)	Export by Estate Office, hashed, with certificate	Corroborative
E8	Corridor CCTV, Block M second floor	Visual identification of the person entering M-204	**High** (7-day overwrite)	Native-format export, hashed	Corroborative
E9	Group-membership change ticket (Aug 2026)	How `mech.labadmin` came to have access to the folder	Low	Copy of the ticketing record	Contextual
E10	Lab staff roster and duty register	Who knew the shared password; who was on duty	Low	Certified photocopy	Contextual

The same items can also be grouped by origin: host-based (E4–E6), server and network (E2, E3), physical security (E7, E8) and documents (E1, E9, E10). Exculpatory material is collected just as carefully as inculpatory material. If the duty register in E10 showed that the card holder was on leave that weekend, it would weaken the case, and the examiner would be obliged to record it.

Q4. Precautions During Collection and Preservation

For the evidence to hold up later, the team must be able to show that it is authentic, complete and unchanged, and that it was handled properly at every step. The precautions below follow the order-of-volatility guidance in RFC 3227 [2] and the ACPO principles [3].

- **Authority before action.** No device is touched until the Principal's written authorisation is on file, and the scope stays limited to college-owned systems. Staff members' personal phones are left out unless the owner agrees.
- **The original is never worked on.** PC07 stays powered off; its SSD is removed and imaged behind a write blocker. The pen drive is never opened on an ordinary computer, because Windows writes to removable media as soon as it is mounted.
- **Most volatile first.** The server Security log and the CCTV recording are collected before anything else because they are overwritten soonest; the disk and pen drive are stable and can wait.
- **A light touch on live systems.** On MECH-FS01 the examiner uses a dedicated forensic account and records each command and its time, so the examiner's own logon events can be separated from the suspect's.
- **Hashes at collection and at every transfer.** SHA-256, with MD5 as a secondary value, is computed when each image or export is made and verified again before analysis.
- **A time reference for every device.** Each device clock is compared with an NTP-synchronised reference and the offset written down. This is how PC07's clock was found to be 4 min 12 s fast.
- **Documentation and packaging.** The lab, the rear of PC07 with the pen drive still inserted, and all labels and serial numbers are photographed. Exhibits go into anti-static bags with numbered tamper-evident seals and are stored in a locked cabinet with an access register.
- **Chain of custody.** Every hand-over records the date and time, who released the item, who received it, why, and the seal number.
- **Independence and confidentiality.** IT staff manage both the server and the shared account, so collection is witnessed and the analysis is done by an examiner from outside the department. Because the exhibits are themselves confidential papers, copies are restricted to the examiner and the committee.

Q5. Establishing the Timeline of the Unauthorized Access

The timeline was built in four steps. First, time-stamped events were gathered from every source: server logs (E2), DC and DHCP logs (E3), PC07 artefacts (E4), pen-drive file entries (E5), RFID (E7) and CCTV (E8). Second, every time was converted to IST. This step needed care, because Windows event logs store UTC, FAT32 stores local time to the

nearest two seconds, and the RFID controller keeps local time. PC07's clock offset of +4 min 12 s, measured at seizure, was confirmed by comparing the DC's Kerberos ticket event (22:41:14) with PC07's own logon record (22:45:27 raw), and every PC07 time was corrected by subtracting it. Third, the events were merged into a single super-timeline and filtered to the window of interest. Finally, no event was accepted as a finding unless a second, independent source supported it.

Time (IST)	Event	Source	Corroborated by
22:31:08	RFID card KIT-ST-0472 opens Block M east door (card issued to a CAD-lab instructor)	E7	E8
22:38:40	CCTV: one person walks along the second-floor corridor and enters M-204	E8	E7
22:41:15	Interactive console logon (Logon Type 2) of `mech.labadmin` on PC07	E4 (corrected)	E3: DC Event 4768 at 22:41:14
22:44:02	SanDisk pen drive (serial 4C530001230614118342) first connected to PC07 and mounted as E:	E4: USBSTOR, setupapi.dev.log, Event 1006	E5: volume serial matches
22:49:31 – 22:55:10	Six question papers read from `\QP-Final\` by `mech.labadmin` from 172.16.40.57	E2: Events 5145/4663	E4: LNK files pointing to `\\MECH-FS01\ExamCell\...`
22:55:48	Six copies created on the pen drive in `E:\ExamPapers\`	E5: FAT32 created times	E4: LNK targets on E:
22:57:20	7-Zip run; `QP_bkp_1209.zip` created on the pen drive	E4: Prefetch `7ZFM.EXE`	E5: ZIP entry and internal timestamps
23:01:37	Loose copies deleted from the pen drive (entries marked 0xE5, still recoverable)	E5	E4: no Recycle Bin use (removable media)
23:09:05	PC07 shut down by the user (Event 1074)	E4	E3: DHCP lease released
23:14:50	RFID card KIT-ST-0472 exits Block M	E7	E8

Interpretation. The logon was made at the console (Type 2). There are no RDP (Type 10) logons and no remote-access software on PC07, so H2 can be set aside, and with no malware found, H3 falls too. The copy is exact: the SHA-256 of `ME3491_Thermal_Engg_SetA.pdf` taken from the ZIP on the pen drive (`b3a5b9de…7e8fe348`) matches the original on the server. What the computer evidence cannot do is name a person, because three people share `mech.labadmin`. For that, the case relies on the RFID card and the CCTV footage, which both point to one CAD-lab instructor. The report therefore says the evidence is **consistent with** that person being present, and recommends a formal interview under the college's disciplinary procedure.

Q6. The Final Forensic Report and Presenting It to a Non-Technical Authority

No.	Section	Contents
1	Title page	Case number, title, examiner's name and designation, date, "Confidential" marking
2	Executive summary	One page in plain language: the allegation, the findings and how certain they are
3	Scope and authorisation	Who authorised the work, which devices and time window were examined, what was excluded
4	Evidence received	Inventory of E1–E10 with descriptions, serial numbers, seal numbers and SHA-256/MD5 values
5	Methodology and tools	Steps followed, tools and versions, write blockers, tool validation, how times were normalised
6	Findings	Numbered factual findings, each tied to the exhibit that supports it
7	Timeline	Table of the reconstructed events, as in Q5
8	Analysis and alternative hypotheses	Why H2 and H3 were rejected and what remains uncertain

No.	Section	Contents
9	Conclusions	Each conclusion with a stated level of confidence ("consistent with", "strongly supports")
10	Limitations and assumptions	Shared account, CCTV resolution, clock offset, data that was not available
11	Recommendations	Reset the six papers; remove `mech.labadmin` from ExamCell-RW; retire shared accounts; enforce MFA and USB control; fix NTP on lab PCs
12	Appendices	Chain-of-custody forms, hash register, log extracts, glossary, examiner's declaration and signature, Section 63 BSA certificate

The Principal and the Examination Committee need to take decisions, not learn forensic tooling, so the briefing starts with the answer and keeps the technical detail in the appendices. Jargon is replaced with plain explanations: a hash value becomes a "digital fingerprint that changes completely if even one character of the file changes", and a write blocker becomes "a one-way valve that lets us read the disk without being able to change it". The timeline table replaces raw log extracts. Facts are kept separate from opinion, and the limits of the evidence are stated plainly, since the shared account means the computer evidence identifies a session rather than a person. An example of the opening paragraph is given below.

> **Sample executive summary, written for the Principal**
>
> On Saturday, 12 September 2026, between 10:49 pm and 10:55 pm, six final question papers for the November examinations were copied from the Mechanical Engineering exam-cell server onto a pen drive in the CAD Laboratory. The copies are exact duplicates of the originals, so the papers must be treated as compromised and reset. The computer records show that a shared lab account was used, which means they cannot on their own identify an individual; the building's door-access and CCTV records point to one member of the CAD-lab staff. We recommend that this person be given the findings and an opportunity to respond before any action is taken.

Q7. Legal and Ethical Considerations in a College Environment

On the legal side, the starting point is **authority**. The college may examine its own servers, its lab PCs and the pen drive found plugged into its PC under its IT policy, but a staff member's personal phone or laptop needs consent or a lawful order. Unauthorised access to and copying of data can attract civil liability under **Section 43 of the Information Technology Act, 2000**, and becomes an offence under **Section 66** when done dishonestly or fraudulently [9]. If the college decides to involve the police, the evidence must already have been handled to court standard. Under **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023**, which replaced Section 65B of the Indian Evidence Act, an electronic record offered in court needs a certificate describing the device and how the record was produced [10]; the hash register and chain-of-custody forms support that certificate. The **Digital Personal Data Protection Act, 2023** requires personal data to be processed only for a stated purpose [11], so unrelated personal files of staff and students found on PC07 are left out of both the analysis and the report. Any disciplinary action must also follow the university's examination-malpractice regulations.

The ethical duties run alongside these. The examiner must stay impartial and look for evidence that clears someone as carefully as evidence that implicates them. Staff who manage the shared account have a conflict of interest and should not analyse the evidence. The exhibits are confidential exam papers and must be protected as such. The person concerned is entitled to see the findings and respond before any decision, in line with natural justice. Finally, the investigation should be proportionate: only what this incident needs is examined, and CCTV footage of unrelated people is not retained.

Reflection

What stood out to me in this case is that recovering the evidence was the easy part; attribution was the hard one. The digital trail is strong and well corroborated, yet a single shared account means the computer can only identify a session, and the case for naming a person rests on door-access and CCTV records. It made the course's emphasis on corroboration, and on fixing basic controls such as shared accounts and USB policy, far more concrete than the textbook description alone.

ASSIGNMENT – 2(2)

Linking a Suspicious File to a User Using File-System Evidence

Unique Scenario: "The Fake Tally Patch". An unknown executable, `TallyBackupHelper.exe`, hidden folders and encrypted archives found on the shared front-office desktop of Nandhini & Co., Chartered Accountants, T. Nagar, Chennai, a fictional firm.

Case No. NCO/DFC/2026/032 · Incident reported: Thu 06 Aug 2026, 10:15 IST · Examiner: M. M. Chandhan Sai

Case Scenario

On 6 August 2026 the antivirus on the front-office desktop **NCO-FRONTDESK** flagged `C:\ProgramData\.syscache\TallyBackupHelper.exe` as an information-stealing Trojan. The `.syscache` folder had its **Hidden** and **System** attributes set, and alongside the executable sat `cfg.dat` and an encoded keystroke log, `kl_0806.log`. Four staff accounts and a shared reception account use this machine, and it holds client GST and income-tax working files. The managing partner wanted two questions answered: which user account created and ran the file, and whether client data had left the firm.

Triage turned up two encrypted items as well: a password-protected 7-Zip archive, `GST_Q1_Clients.7z` (AES-256), in `C:\Users\karthik.s\Documents\Personal\`, and a 2 GB file, `notes.hc`, with no recognisable header, in a hidden folder `C:\Users\karthik.s\AppData\Local\Temp\~cache\`. Not surprisingly, early suspicion settled on the account `karthik.s`, belonging to an article assistant.

Why I chose this scenario. Malware disguised as an accounting-software update is a realistic threat to small Indian firms that run Tally, and a shared office desktop is exactly the setting in which "which user did this?" becomes hard. The case also contains both hidden and encrypted files, so each question in the assignment has something real to work on.

Account	Role	Notes
`anitha.r`	Office manager	Local administrator
`karthik.s`	Article assistant	Standard user; the account under suspicion
`meena.p`	Accounts clerk	Standard user
`nco.admin`	External IT vendor	Local administrator; installed AnyDesk for remote support
`reception`	Shared front-desk account	Standard user with a **blank password**

Assumptions

- All names, paths, hashes and values are fictional. The firm owns the desktop, and staff signed an IT policy permitting inspection of office systems.
- The IT vendor shut the desktop down normally on 6 Aug at 10:40, so no memory capture was possible.
- ILookIX is the main analysis tool, and its steps describe the intended workflow rather than an operation actually performed.

Q1. Hardware and Software Components to Document Before Analysis

The environment is documented before any analysis for two reasons: it makes the examination repeatable, and it explains how the artefacts found later came to exist. A clock offset or a remote-access tool noticed at this stage can change how every later finding is read. The **system inventory** recorded for NCO-FRONTDESK is shown below.

Component	Recorded detail	Why it matters
Make / model / service tag	Dell OptiPlex 3080 SFF, tag 7QX2K53	Uniquely identifies the exhibit
CPU / RAM	Intel Core i5-10500, 8 GB DDR4	Context for hibernation and page-file size
Primary storage	512 GB SATA SSD, Crucial MX500, S/N 2214E61A0B3C	TRIM may permanently erase deleted data
Secondary storage	1 TB Seagate HDD (D:, nightly Tally backups)	May hold older copies of files

Component	Recorded detail	Why it matters
Firmware and clock	BIOS 2.21.0; TPM 2.0 present; RTC **+37 s** against reference	The offset corrects every timestamp
Network	Realtek GbE, MAC 8C:EC:4B:5A:21:9F, static IP 192.168.1.23	Links router and firewall logs to this PC
Operating system	Windows 10 Pro 22H2, build 19045.4651, installed 14 Jan 2026	Any file "created" before this date is suspect
Time zone	India Standard Time (UTC+05:30)	Needed to convert NTFS UTC times
File systems	C: NTFS, D: NTFS; BitLocker **off**; 2 Volume Shadow Copies	Determines which artefacts exist
User accounts	5 accounts, SIDs ending -1001 to -1005	Maps SIDs in artefacts to people
Security software	Quick Heal Total Security; Windows Defender disabled	AV logs give the detection time
Remote access	**AnyDesk 8.0.9**, installed by the IT vendor	Someone could have used the PC remotely
Key applications	Tally Prime 4.1, MS Office 2019, Chrome 127, Edge, 7-Zip 23.01, VeraCrypt 1.26.7 (portable)	Each leaves its own artefacts
Audit settings	Logon/logoff auditing on; process creation (4688) **off**	Explains gaps in the event log
Peripherals	HP LaserJet M1136, webcam, USB keyboard	Possible print or USB activity

Q2. How Storage, the OS, the File System and Metadata Help Locate Evidence

Storage devices. The SSD holds the operating system and the user profiles, but because of TRIM, deleted files on it are often gone for good. The secondary HDD has no TRIM and receives nightly Tally backups, so it may still hold earlier versions of client files and deleted remnants. Both drives are imaged in full so that unallocated and slack space can be searched as well.

Operating system. Windows organises a great deal of evidence by user. Each account has its own profile folder (`C:\Users\<name>`) and its own registry hives (`NTUSER.DAT`, `UsrClass.dat`), so artefacts such as UserAssist, ShellBags, RecentDocs and Chrome history show *which account* acted. System-wide sources, such as Prefetch, the event logs and the SYSTEM and SOFTWARE hives, show *what happened on the machine*, but usually not who did it.

File system (NTFS). Every file has a Master File Table (`$MFT`) record carrying two sets of timestamps [5]. The `$STANDARD_INFORMATION` set is easy for programs to change; the `$FILE_NAME` set is updated only by the operating system. When the two disagree, timestamp tampering is the likely explanation. NTFS also records the **owner SID** of each file, keeps **Alternate Data Streams** such as `Zone.Identifier` (which stores the download URL), logs creations, renames and deletions in the change journal `$UsnJrnl`, and gives each user a Recycle Bin folder named after their SID.

Metadata. Some metadata travels inside the file itself. The executable's PE header gives a compile time and shows that the file is unsigned; Office documents record an author and a "last modified by" name; ZIP and 7-Zip archives list their internal file names and times even when the contents are encrypted.

Layer	Example from this case	What it located
Storage	Secondary HDD (D:) backups	Older client files for comparison
OS	`karthik.s` NTUSER.DAT → UserAssist	Execution of the EXE by that account
File system	`$MFT` record of `TallyBackupHelper.exe`	`$SI` created 02 Nov 2025 vs `$FN` created 03 Aug 2026, i.e. timestomping
File system	`Zone.Identifier` ADS on the downloaded ZIP	Download URL `hxxps://tally-prime-update[.]co/download`
Metadata	PE header of the EXE	Compiled 28 Jul 2026, unsigned, not a genuine Tally file

Q3. How Hidden and Encrypted Files Affect the Investigation

Hidden files do not actually stop a forensic tool, which reads the file system directly and ignores the Hidden attribute. What they do is signal intent and change how the evidence is read. Data can be hidden in several ways: by the Hidden and System attributes (as with `.syscache`), by dot-prefixed folder names, in Alternate Data Streams, behind a misleading extension such as an EXE renamed to `.dat`, or simply by being placed in folders nobody browses, such as `ProgramData` or `AppData\Local\Temp`. To find all of these, the examiner relies on **signature analysis**, which checks file headers rather than extensions, and on an **attribute filter**.

Encrypted files are the more serious obstacle, because their contents cannot be read without the key. Two such items appear here. `GST_Q1_Clients.7z` is AES-256 encrypted, but since header encryption was not used, 7-Zip leaves the **file list visible**: it names 41 client GST return files, which tells us what was archived without decrypting anything. `notes.hc` has no header and a uniform entropy of 7.99 bits per byte, the typical profile of a **VeraCrypt container**, and Prefetch and UserAssist show that VeraCrypt portable was run from `karthik.s`'s Downloads folder.

Evidence-handling considerations. Password guessing is never attempted on the original, only on verified copies, and each encrypted file is hashed and its location and attributes recorded before any attempt at decryption. The image is first searched for keys that are lawfully available: browser password stores, notes, password-manager files, and memory remnants in `hiberfil.sys` and `pagefile.sys`. The password itself should be requested through proper process; compelled disclosure raises self-incrimination questions under Article 20(3) of the Constitution of India, so legal advice is needed. If consent is not forthcoming, the 7-Zip hash can be extracted and attacked offline on a copy, with the tool and word list documented. Anything decrypted is treated as **derived evidence**, with a record of how it was produced and its own hash.

It is worth stressing that hidden or encrypted does not mean guilty. `karthik.s` gave the VeraCrypt password voluntarily, and the container turned out to hold only his CA Intermediate study notes. It was **excluded** from the report as both out of scope and private.

Q4. Evidence-Linkage Matrix

The matrix brings together file names, paths, timestamps, account SIDs and metadata so that each link between `karthik.s` and the suspicious file can be checked on its own.

#	Artefact	Path / source	Time (IST)	Account	Key detail	Link type
L1	Logon event 4624, Type 2	Security.evtx	03 Aug 18:47:02	karthik.s	Interactive console logon	Supporting
L2	Chrome download record	`karthik.s\...\Chrome\User Data\Default\History` (downloads table)	03 Aug 18:52:10	karthik.s	`TallyBackupHelper_setup.zip`; referrer: Gmail web	**Direct**
L3	Zone.Identifier ADS	`C:\Users\karthik.s\Downloads\TallyBackupHelper_setup.zip:Zone.Identifier`	03 Aug 18:52:11	karthik.s (profile)	ZoneId=3, HostUrl = fake Tally site	Supporting
L4	`$MFT` record of EXE	`C:\ProgramData\.syscache\TallyBackupHelper.exe`	`$FN` 03 Aug 18:53:41	Owner SID …-1004 (karthik.s)	`$SI` shows 02 Nov 2025, i.e. timestomped	**Direct**
L5	UserAssist (ROT13)	karthik.s `NTUSER.DAT`	03 Aug 18:53:44	karthik.s	Run count 1, launched via Explorer	**Direct**
L6	Prefetch	`C:\Windows\Prefetch\TALLYBACKUPHELPER.EXE-3F1A2C9B.pf`	First 03 Aug 18:53:44; last 06 Aug 09:58	Not recorded	9 runs	Supporting
L7	Run key persistence	karthik.s `NTUSER.DAT` → `...\CurrentVersion\Run\TallyBkp`	Key written 03 Aug 18:53:52	karthik.s	Starts the EXE at each karthik.s logon	**Direct**
L8	ShellBags	karthik.s `UsrClass.dat`	03 Aug 18:55:20	karthik.s	Explorer opened `C:\ProgramData\.syscache`	**Direct**

#	Artefact	Path / source	Time (IST)	Account	Key detail	Link type
L9	7-Zip Jump List	karthik.s `AutomaticDestinations`	05 Aug 17:20:33	karthik.s	`GST_Q1_Clients.7z` created	Contextual
L10	AnyDesk `ad_svc.trace`	`C:\ProgramData\AnyDesk\`	04 Aug 13:08–13:31	Unknown remote ID	Incoming session; no IT ticket	Contextual (exculpatory)
L11	Leave register	HR record	04 Aug (full day)	—	karthik.s on approved leave	Contextual (exculpatory)

Q5. Acquiring the Evidence and Analysing It with ILookIX

ILookIX is a forensic analysis suite that reads common image formats, including the ASB images produced by its companion imager IXImager. It supports a wide range of file systems (FAT, NTFS, HFS+ and ext among them), can break down Volume Shadow Copies, and can recover deleted files together with their metadata [8]. The intended workflow for this case has ten steps.

1. **Acquire.** The SSD and HDD are removed and each is imaged through a hardware write blocker with IXImager into an ASB container, recording source and image hashes (SSD image SHA-256: `c5d55aaa…2f45f38b`).
2. **Create the case.** The case number and examiner are entered and the **display time zone is set to IST**, with a note that the underlying NTFS values are UTC. Two hash libraries are loaded: a known-good set to hide standard Windows and Office files, and a known-bad set of malware hashes.
3. **Import and verify.** Both images are added, and the hash ILookIX computes must match the acquisition hash before any processing begins.
4. **Process.** ILookIX identifies partitions and volumes, parses NTFS, breaks down the two Volume Shadow Copies and recovers deleted files. It then runs **signature analysis** to catch renamed files, **hash analysis**, and **entropy analysis** to flag encrypted containers such as `notes.hc`. Archives are expanded so the 7-Zip file list becomes searchable.
5. **Filter and reduce.** Known-good files are hidden, and the remainder is filtered by user profile and SID, by Hidden and System attributes, and by the date range 01–06 Aug 2026.
6. **Analyse user activity.** Each user's `NTUSER.DAT` and `UsrClass.dat` are parsed (UserAssist, Run keys, ShellBags, RecentDocs), together with the SAM hive for account SIDs and last logons, Prefetch, LNK files, Jump Lists and the event logs.
7. **Analyse web artefacts.** Chrome and Edge history, downloads and cache are extracted for every profile. This step recovered the cached Gmail page showing the phishing e-mail titled "Mandatory Tally Prime backup patch".
8. **Search.** A keyword search covers `TallyBackupHelper`, `syscache` and `tally-prime-update`, and a regex search for GSTIN numbers (`\d{2}[A-Z]{5}\d{4}[A-Z][1-9A-Z]Z[0-9A-Z]`) checks whether client identifiers appear in `kl_0806.log` or in outbound data.
9. **Build the timeline and bookmark.** Events from 03–06 Aug are placed on a timeline, and artefacts L1–L11 are bookmarked with notes marking each as direct or supporting.
10. **Report and preserve.** Bookmarked items are exported with their metadata and hashes, the report is generated, and the ILookIX case file is archived with the images.

Q6. Direct Links versus Supporting or Contextual Artefacts

A **direct** artefact either lives inside a particular user's profile or records that user's SID as the actor, so it ties the *account* to a specific action. A **supporting** or **contextual** artefact shows that something happened on the system, or describes the surrounding circumstances, but cannot by itself say which account was responsible.

Aspect	Direct link	Supporting / contextual
Definition	Records the action within a specific account's data	Records system-wide activity or surrounding circumstances
Examples here	UserAssist and Run key in karthik.s `NTUSER.DAT` (L5, L7); Chrome download in his profile (L2); owner SID on the EXE (L4); ShellBags (L8)	Prefetch (L6) shows the EXE ran, not who ran it; Zone.Identifier (L3) shows where it came from; the logon event (L1); the AnyDesk log (L10); the leave register (L11)
Strength	Strong for the **account**	Strengthens the direct links when it agrees with them, weakens them when it does not

Aspect	Direct link	Supporting / contextual
Limitation	Still links an *account*, not a *person*, if credentials are shared	Cannot support attribution on its own

This distinction matters more than it first appears. Even the strongest direct artefact proves only that the **karthik.s account** acted. Linking a **person** requires evidence that only that person could have been using the account at the time, such as physical presence, CCTV, an interview, or exclusive knowledge of the password.

Q7. Limitations and Alternative Explanations

Before attributing the activity to the person behind `karthik.s`, six alternative explanations have to be weighed:

1. **A shared or known password.** A sticky note bearing his password was photographed under the keyboard during collection, so anyone in the office could have logged in as him.
2. **Remote access through AnyDesk.** A session from an unknown remote ID ran on 04 Aug from 13:08 to 13:31, while karthik.s was on approved leave, and coincides with a logon under his account. At least some of the activity under his name was therefore **not** his.
3. **The malware acting on its own.** After the first run, the program created `cfg.dat` and `kl_0806.log` itself and relaunched at each logon. Files created by malware inherit the logged-on user's context, so their ownership says nothing about human action.
4. **Timestamp manipulation and clock error.** The installer back-dated the EXE's `$SI` times, and the BIOS clock ran 37 seconds fast. Only `$FN` times and corrected values can be relied on.
5. **A victim rather than an author.** The browser cache shows a convincing phishing e-mail. Downloading and running the file is entirely consistent with a user being deceived, so the action does not prove malicious intent.
6. **The blank-password reception account.** Another person could have used it to stage files, although nothing in the `reception` profile relates to the executable.

Conclusion

The evidence shows that the **karthik.s account** downloaded `TallyBackupHelper_setup.zip` from a fake Tally update site on 03 Aug 2026 at 18:52 IST, after following a link in a Gmail message, and launched the extracted executable through Explorer at 18:53:44. The installer back-dated its own timestamps, hid itself in `C:\ProgramData\.syscache` and set a Run key so it would start at every karthik.s logon. The keystroke log contains typed text that includes client GSTINs, so client data must be treated as exposed.

The link to the account is strong, but the link to the person is not conclusive: his password was written down beside the PC, and an unexplained AnyDesk session used the account on 04 Aug while he was on leave. The 7-Zip archive of client GST files breaches the firm's data-handling policy but is a separate matter, and the VeraCrypt container was unrelated. I would recommend resetting every password, removing AnyDesk or restricting it to approved IDs, enabling process-creation auditing, training staff on phishing, and notifying affected clients.

Reflection

This case taught me to be careful with the word "user". Nearly every artefact pointed at one account, and it would have been easy to stop there. Only when the AnyDesk log and the leave register were set against the timeline did it become clear that the account and the person are not the same thing, which is precisely the point of separating direct from supporting evidence.

ASSIGNMENT – 2(3)

Forensic Imaging and Safekeeping of a Suspect Storage Device

Unique Scenario: "The Departing Firmware Engineer". Seizure and IXImager imaging of a laptop and USB drive in a suspected theft of battery-management firmware from VoltNova Energy Systems Pvt. Ltd., Guindy Industrial Estate, Chennai, a fictional EV battery start-up.

Case No. VNE/DFC/2026/005 · Seizure: Thu 24 Sep 2026, 10:42 IST · Examiner: M. M. Chandhan Sai

Case Scenario

VoltNova's data-loss-prevention (DLP) system raised an alert on **22 Sep 2026 at 13:36:10 UTC (19:06:10 IST)**: 4.2 GB had been written from a senior firmware engineer's company laptop to a removable USB device, which the DLP agent identified as a **Kingston DataTraveler Exodia 64 GB**. The engineer had resigned two weeks earlier and was due to join a competitor. The data at risk is the source code of the company's **Battery Management System (BMS) firmware v4.2**.

On 24 Sep, with the CEO's written authorisation and the HR Manager present, the laptop was found at desk R-14 **powered on and screen-locked**, with the same Kingston drive still inserted. Its internal drive is protected by **BitLocker**, and the recovery key is escrowed in the company's directory service.

Why I chose this scenario. Insider data theft by an employee who is about to leave is one of the most common cases in corporate forensics. Finding the laptop still running, encrypted, and with the USB drive attached forces real decisions at the scene, and a quick-formatted drive gives a genuine reason to use data carving.

Exhibit	Description
E1, laptop	Lenovo ThinkPad T14 Gen 4, S/N PF4K2X9M (fictional); Windows 11 Pro 23H2; BitLocker on C:
E1a, internal drive	Samsung PM9A1 512 GB NVMe M.2 SSD, S/N S6XNNF0W812345 (fictional)
E2, USB drive	Kingston DataTraveler Exodia 64 GB, S/N 60A44C413B9BF2A0 (fictional), exFAT
E3, power adapter	Lenovo 65 W USB-C adapter, seized so the laptop can be kept charged if needed
Reference	DLP alert export (UTC) and HR resignation record

Assumptions

- All names, serial numbers, hashes and values are fictional. VoltNova owns both the laptop and the USB drive, and the employment contract permits inspection.
- IXImager is used for imaging and the ASB container for storage, as the case specifies. Steps describe the intended workflow.
- The laptop's M.2 SSD is removable, and the lab has a write-blocked NVMe bridge.

Q1. Collecting and Transporting the Laptop and USB Drive

Because the laptop was **running**, the procedure has to start with a decision about the live system before anything is physically moved. Figure 3.1 summarises the full sequence, and the steps are as follows.

1. **Secure the scene.** Keep the employee and other staff away from desk R-14 and record who was present. Nobody touches the keyboard or mouse.
2. **Photograph and document.** Photograph the lock screen and the time shown on it, the desk, every port including the inserted USB drive, and all cables and labels. Record makes, models and serial numbers.
3. **Record the clock.** The lock screen read 10:44:51 against the investigator's NTP-synchronised reference of 10:42:30, so the laptop is **2 min 21 s fast**.
4. **Decide on the live system.** The drive is BitLocker-encrypted, but since the recovery key is escrowed, the disk can be decrypted later and memory capture is not strictly needed for that purpose. RAM may still hold evidence of running processes and recent USB activity, however, so a **memory capture** is taken first from the examiner's validated USB toolkit, and the toolkit's own footprint is documented. Had the key not been available, a live acquisition of the unlocked volume would have been essential.
5. **Isolate from networks.** Airplane mode is used if it can be enabled without logging in; otherwise the laptop goes into a Faraday bag immediately after shutdown, so it cannot be wiped or synchronised remotely.

6. **Power down.** A hard power-off is performed by holding the power button, because a normal shutdown could run updates or clean-up scripts. The time is recorded (10:58).
7. **Remove the USB drive** only after power-off, so that no "eject" writes occur, and photograph it with its label and serial number.
8. **Package.** The USB drive goes into an anti-static bag and the laptop into a padded anti-static Faraday bag. Each is sealed with a numbered tamper-evident seal (VNE-0051 for the laptop, VNE-0052 for the USB drive), signed across the seal.
9. **Label and log.** Each label carries the case number, exhibit number, description, date and time, and the seizing officer's name, and the chain-of-custody form (Q8) is started.
10. **Transport.** The exhibits travel in a locked, padded case in the investigator's personal custody, away from heat, magnets, moisture and shocks, directly to the lab with no unattended stops.
11. **Hand over** to the lab evidence custodian, who checks the seals, signs the custody form and locks the exhibits away.

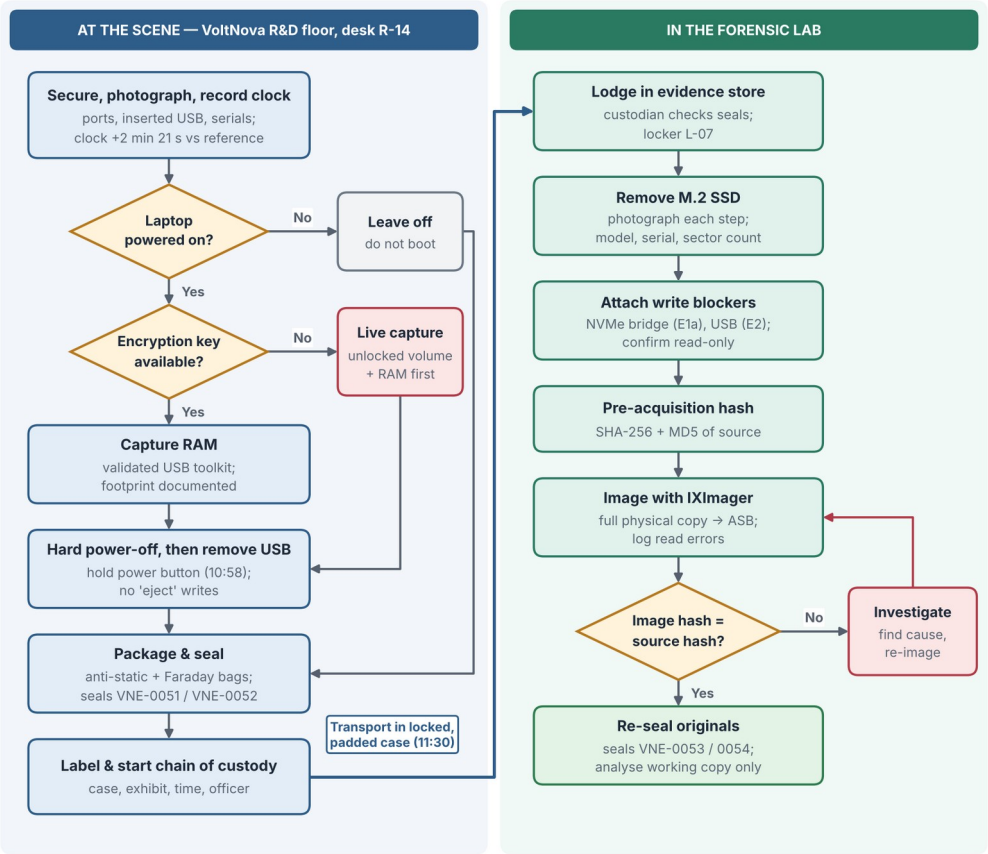


Figure 3.1: Acquisition flowchart, from seizure of a running laptop to verified forensic images

Q2. Why Physical Acquisition, and Why We Analyse the Image

Physical acquisition copies a storage device bit for bit, sector by sector, from the first sector to the last. A logical copy captures only the visible files; a physical image also captures deleted files still sitting in unallocated space, file and volume slack, partition tables, boot records, unpartitioned areas, and file-system structures such as the `$MFT` or the exFAT allocation bitmap and directory entries. For the Kingston drive this is the whole point: the suspect is believed to have quick-formatted it, and the old data survives only in the unallocated space that a physical image includes.

Analysis is then done on the image rather than the original for several connected reasons. Simply mounting or browsing the original makes the operating system write to it (journal updates, access times, indexing), whereas the image can be examined without any risk to the source. The image's hash can be compared with the original's to prove it is an exact copy, and because the original stays sealed, the defence or a second expert can re-examine the same data and should reach the same results. Risky operations such as carving, decryption and mounting can be run on copies, with a fresh copy made from the master if one is damaged. Several examiners can work in parallel, and the evidence remains preserved even if the original hardware later fails.

Q3. An IXImager Imaging Workflow

IXImager is the imaging component of the ILook suite. It runs on a Linux base independent of Windows and writes images in authenticated proprietary formats (compressed, uncompressed and encrypted) stored as ASB files, which can be converted to raw format for other tools [8]. When it was tested under the NIST CFTT programme, IXImager acquired all visible and hidden sectors accurately and replaced unreadable sectors with benign content [6].

Step	Stage	Action for E1a (NVMe SSD) and E2 (USB)
1	Prepare	Use a validated forensic workstation; wipe and verify the destination drive; record the IXImager version and workstation ID
2	Source identification	Remove the M.2 SSD from the laptop, photographing each step. Record model, serial number, firmware, capacity and the total sector count reported by the device (E1a: 1,000,215,216 sectors × 512 B; E2: 121,110,528 sectors) and check them against the label
3	Write protection	Connect E1a through a hardware NVMe write blocker and E2 through a USB 3.0 write blocker. Confirm the write-block LED and test that the OS cannot mount the source read-write
4	Pre-acquisition hash	Compute SHA-256 and MD5 of each source through the write blocker
5	Imaging	Select the source in IXImager and choose ASB output (compressed authenticated format for E1a; uncompressed for E2 to simplify carving). Set the segment size, enter case, exhibit and examiner details, and run a full physical acquisition, logging any read errors or bad sectors
6	Verification	IXImager hashes the data stream as it is acquired; the finished image is then re-read and hashed, and both values are compared with the pre-acquisition source hash
7	Post-acquisition hash	Re-hash the source through the write blocker to prove it did not change during imaging
8	Documentation	Save the IXImager log; record start and end times, all hashes, any errors, and the write blocker's model and serial; photograph the setup; sign the acquisition worksheet
9	Working copy and decryption	Make a working copy of the ASB image and unlock BitLocker **on the copy** with the escrowed recovery key, recording the key ID used. The master image stays sealed
10	Re-seal originals	Re-install the SSD, re-seal both exhibits with new seals (VNE-0053, VNE-0054) and return them to the locker

Q4. How Hash Values Demonstrate Integrity

A cryptographic hash function reduces any amount of data to a fixed-length value. Change a single bit of the input and the output changes completely, and finding two different inputs with the same SHA-256 value is computationally infeasible. Matching hashes therefore mean matching data. SHA-256 is used as the primary algorithm, and MD5 is recorded as a secondary value for compatibility with older tools, since MD5 on its own is no longer considered collision-resistant [4].

Stage	SHA-256 (E1a, NVMe SSD)	Result
Source before imaging (25 Sep 10:02)	`382737eb45e656241d33174d321533c3 50272beb46fa11fa6186df41d84e56c8`	Baseline
Acquired data stream (IXImager)	`382737eb45e656241d33174d321533c3 50272beb46fa11fa6186df41d84e56c8`	**Match**
Source after imaging (25 Sep 11:47)	`382737eb…41d84e56c8`	**Match**: source unchanged
Working copy before analysis (26 Sep)	`382737eb…41d84e56c8`	**Match**: copy is exact

Stage	SHA-256 (E2, USB)	Result
Source before imaging	`6a44b74d6ef76f5188a7e90cc84af074 b2304acf094b17cfbb119404809df339`	Baseline
Verified image	`6a44b74d…809df339`	**Match**
Source after imaging	`6a44b74d…809df339`	**Match**

One distinction is easy to miss. The hash of the *data inside* the image must equal the hash of the source, but the ASB container file has a different hash (here `e5c9ec5f…c411f7c9`), because it also holds metadata and compressed blocks. That container hash is recorded separately and checked during storage. If a mismatch ever appears, the image is set aside until the cause is found and documented, and a new image is taken from the original if necessary.

Q5. The Role of the ASB Container and What Must Accompany the Evidence

An **evidence container** keeps the acquired image together with the record of how, when and by whom it was acquired, so that the image and its documentation cannot drift apart. The ILook/IXImager **ASB** format is such a container. It offers authenticated storage, with mechanisms that detect any change between the source and the stored image; optional compression to save space; optional encryption for secure transport and archiving; and logging of the acquisition itself [7], [8]. ILookIX reads ASB images directly, and IXImager can convert them to raw format when another tool needs them.

Category	Information that should accompany the evidence
Case identity	Case number VNE/DFC/2026/005, exhibit numbers E1a/E2, short description
Examiner	Name, designation, organisation, signature or ID
Source device	Make, model, serial number, firmware, capacity, sector count, interface, photographs
Acquisition details	Tool and version (IXImager), workstation ID, write blocker model and serial, start and end times with **time zone**, format chosen (compressed or encrypted)
Integrity values	Source hash before and after, image hash, container hash, algorithms used
Errors	Bad or unreadable sectors and how they were handled
Clock information	Device clock against reference (+2 min 21 s) and the device's time-zone setting
Encryption	BitLocker present; recovery key ID used for the working copy (the key itself is stored separately and securely)
Custody	Seal numbers, chain-of-custody reference, storage location
Notes	Live-capture actions taken before shutdown and any departures from standard procedure

Q6. Data Carving and When It Is Needed

Data carving recovers files from raw data **without relying on file-system metadata** [5]. The examiner scans the image for known file signatures, a header marking where a file begins and, where one exists, a footer or length field marking where it ends, and extracts the bytes in between.

File type	Header	Footer / end marker
ZIP	`50 4B 03 04` ("PK..")	End-of-central-directory `50 4B 05 06`
PDF	`25 50 44 46` ("%PDF")	`%%EOF`
JPEG	`FF D8 FF`	`FF D9`
Intel HEX (firmware)	Lines starting with `:`	`:00000001FF` end-of-file record

Carving becomes necessary whenever the file system can no longer point to the data. That happens when a volume has been **quick-formatted**, leaving the data in place but resetting the directory and allocation bitmap; when files have been **deleted** and their directory entries overwritten or reused; when the partition table or boot sector is **damaged or deliberately wiped**; when the data lies in **unallocated space, slack space, page or hibernation files**, or memory dumps that have no file-system structure; and when the file system is **unknown or unsupported** by the tool.

In this case, the exFAT volume on E2 was created on **23 Sep 2026 at 08:14**, the day after the DLP alert, with a new volume serial number, and it now holds only two music files. Carving its unallocated space recovered a 3.9 GB ZIP archive whose central directory lists the `bms_core/` source files of firmware v4.2, along with a PDF whose internal title is "Cell Balancing Specification Rev C". Because the data had been written in one session to an almost empty drive, the files were contiguous and carved cleanly.

Carving has real limits, though. Carved files come back **without their original names, paths or file-system timestamps**; fragmented files may be incomplete; and false positives are possible. On the NVMe SSD (E1a), TRIM usually zeroes

deleted data, so carving there is unlikely to help. Every carved result is therefore verified by opening it in a viewer, checking its internal structure, and comparing its hash with the company's repository copy where one exists.

Q7. Date and Time Problems in Digital Evidence

Timestamps are the backbone of most cases, yet they come from different clocks, formats and time zones. This case shows the typical problems clearly.

Problem	How it appears in this case
Clock drift	The laptop's RTC was **2 min 21 s fast** at seizure
Wrong time-zone setting	The laptop was still set to **(UTC+04:00) Abu Dhabi, Muscat** after a trip to a Dubai trade show, so local times shown by Windows are 1 h 30 min behind IST
UTC vs local storage	NTFS stores UTC; exFAT stores local time with an optional UTC-offset field; ZIP stores DOS local time with **no zone** and 2-second precision
Different log sources	DLP server logs are in UTC; HR records are in IST
Daylight saving	Not used in India or the UAE, but relevant whenever evidence crosses regions
Timestamp manipulation	Tools can alter `$STANDARD_INFORMATION` times, so they are compared with `$FILE_NAME`
Resolution differences	NTFS 100 ns; FAT and ZIP 2 s; most logs 1 s

A worked example. The ZIP carved from E2 shows internal file times of **22 Sep 2026, 17:36** in the laptop's local time (UTC+4). Converting gives 17:36 − 4:00 = 13:36 UTC = **19:06 IST**, which matches the DLP alert at 13:36:10 UTC. Had the wrong time-zone setting gone unrecorded, the files would appear to have been archived an hour and a half before the alert.

To keep such uncertainties under control, each device's clock is recorded against a reference at seizure, with the size and sign of the offset, and the configured time zone is taken from both the lock screen and the registry (`SYSTEM\CurrentControlSet\Control\TimeZoneInformation`). Every timestamp in the report states its source, storage format (UTC or local) and precision. All times are normalised to IST in a single documented step, with raw and corrected values shown side by side; tolerance windows are given instead of false precision (for example ±2 s for ZIP times); and no important conclusion rests on a single timestamp without an independent source, such as the DLP log, to back it up.

Q8. Chain-of-Custody Form

Chain-of-custody form for Case VNE/DFC/2026/005, Exhibits E1 (laptop with E1a SSD) and E2 (USB drive). Both parties sign every row; any broken seal is photographed and explained; and the form travels with the exhibit for the life of the case.

#	Date / time (IST)	Action / purpose	Released by	Received by	Location	Seal / condition
1	24 Sep 2026 10:42	Seizure at desk R-14; memory captured; hard power-off 10:58	—	A. Prakash, Investigator (witness: HR Manager)	VoltNova R&D floor	VNE-0051 (E1), VNE-0052 (E2) applied, intact
2	24 Sep 11:30	Transport to forensic lab in locked padded case	A. Prakash	A. Prakash (in personal custody)	Vehicle TN-09-AB-1234	Seals intact
3	24 Sep 13:05	Lodged in evidence store	A. Prakash	M. Sheela, Evidence Custodian	Lab locker L-07	Seals checked, intact
4	25 Sep 09:15	Released for imaging	M. Sheela	M. M. Chandhan Sai, Examiner	Imaging bench IB-2	Seals broken by examiner, photographed
5	25 Sep 12:40	Imaging and verification complete; returned	M. M. Chandhan Sai	M. Sheela	Lab locker L-07	Re-sealed VNE-0053 (E1), VNE-0054 (E2)

#	Date / time (IST)	Action / purpose	Released by	Received by	Location	Seal / condition
6	26 Sep 10:00	Working copy of ASB images issued (originals not removed)	M. Sheela (evidence server)	M. M. Chandhan Sai	Analysis workstation AW-4	Image hash verified, match
7	03 Oct 16:30	Examination complete; master images archived	M. M. Chandhan Sai	M. Sheela	Evidence server and offline backup	Hashes re-verified, match

Reflection

Before this assignment I thought of imaging as a single "copy the disk" step. Working through a running, encrypted laptop showed me how many decisions come before the copy even starts, and how one overlooked detail, here a time-zone setting left over from a business trip, could shift the whole timeline by ninety minutes. Documenting the clock at seizure now seems to me as important as the hash itself.

ASSIGNMENT – 2(4)

Recovery of Deleted Evidence from Volume Shadow Copies

Unique Scenario: "The Vanishing Tender Sheet". Recovery of a deleted port-tender pricing workbook and cleared browser history from Volume Shadow Copies on a pricing analyst's laptop at Coromandel Freight & Logistics Pvt. Ltd., Ennore, Chennai, a fictional company.

Case No. CFL/DFC/2026/011 · Image received: Sat 19 Sep 2026 · Examiner: M. M. Chandhan Sai

Case Scenario

Coromandel Freight & Logistics bid for a cargo-handling tender at Ennore Port. When the results came out on **18 Sep 2026**, a competitor had undercut Coromandel's rates by just 0.4% on every line item, which is too close to be coincidence. Suspicion fell on a pricing analyst, **Ms. L. Revathi** (account `revathi.l`), who had prepared the rate workbook `Ennore_Tender_Rates.xlsx`. When her laptop was collected on 19 Sep, the workbook was **missing**, the Recycle Bin was empty, and Microsoft Edge's history was almost blank.

A forensic image (E01, SHA-256 `305622d5…b9008267`, verified) of her Dell Latitude 5440 running **Windows 11 Pro 23H2** was provided for analysis. Its `System Volume Information` folder contains **four Volume Shadow Copies (VSCs)**, created automatically by System Restore and Windows Update.

Why I chose this scenario. Volume Shadow Copies are a good example of the operating system quietly preserving evidence that a user believes is gone. A deleted file combined with cleared browser history is the classic anti-forensics pattern, and a tender leak gives the timeline real meaning, since the order of events is the whole case.

VSC	Created (IST)	Trigger
SC1	26 Aug 2026 02:00	Scheduled restore point
SC2	03 Sep 2026 02:00	Scheduled restore point
SC3	**11 Sep 2026 13:05**	Windows cumulative update installation
SC4	16 Sep 2026 02:00	Scheduled restore point

Assumptions

- All names, hashes and values are fictional. The laptop is company property, and the employee consented to inspection under company policy.
- The final workbook was e-mailed internally to the Commercial Manager on 08 Sep at 18:05, and that copy survives on the company mail server as an independent reference.
- Tool steps describe the intended workflow, with ILookIX used for VSC breakdown and a second tool for verification.

Q1. Selecting and Categorising Evidence Before Detailed Analysis

A full disk image holds hundreds of thousands of files, most of them irrelevant. Before any detailed analysis, the investigator has to **select** what matters and **categorise** it, so that effort goes where it counts and private, unrelated data is left untouched.

The process begins with the questions the case must answer: did the workbook exist, who created and changed it, was it copied or sent out, when and how was it deleted, and was there an attempt to hide activity? These questions set the scope, here the window 25 Aug – 19 Sep 2026, the `revathi.l` profile, and the relevant file types (Office documents, archives, browser data, e-mail, LNK files, USB artefacts and event logs). Data is then reduced by removing known operating-system and application files with a known-good hash set such as NSRL, and by setting aside personal media such as music and photos that cannot be relevant. The sources of deleted and historical data are listed next: the Recycle Bin, `$UsnJrnl`, `$LogFile`, unallocated space and the four VSCs. Finally, the remaining items are categorised using the scheme in Q2 and given priorities. Every selection decision, including what was excluded and why, is recorded so that the reduction is transparent and can be repeated.

Q2. Evidence Classification Scheme

Category	Description	Examples in this case	Priority
User-created files	Documents created or edited by the user	`Ennore_Tender_Rates.xlsx` (in SC2 and SC3), `tender_notes.docx`, draft e-mails	High
System / application files	OS and program files, plus system-generated logs and databases	Excluded: Windows binaries and Office program files (known-good hashes). Retained: event logs, registry hives, `$MFT`, `$UsnJrnl`, VSC catalog	Low (binaries) / High (logs)
Potentially suspicious files	Items suggesting concealment, exfiltration or misuse	Edge "Clear browsing data – All time" record in `Preferences`; `hr_policy.pdf` whose header is actually ZIP/XLSX (signature mismatch); USB device history	High
Deleted / recovered artefacts	Items no longer visible on the live volume but recovered	Workbook from SC3; Edge `History` database from SC3; `$UsnJrnl` records of the deletion; `$I`/`$R` Recycle Bin names	High
Supporting / contextual evidence	Information that explains or corroborates	VSC creation times, Windows Update log, Office MRU list, logon events, internal e-mail copy, tender result date	Medium

Q3. How Event Analysis Helps Reconstruct a Sequence of Activities

Event Analysis brings time-stamped events from many sources (the file system, registry, event logs, browser databases, LNK files and the VSCs themselves) into a single ordered sequence and then looks at how those events relate to one another. Its first contribution is simply ordering: once "file modified", "USB connected", "web-mail upload page visited" and "file deleted" sit on the same time axis, it becomes clear what happened first.

Its more powerful contribution is revealing **clusters**. In this case, five events fall within five minutes on 11 Sep, between 18:44 and 18:49: the workbook is deleted, the Recycle Bin is emptied, and Edge's history is cleared. Bursts like this point to deliberate clean-up rather than routine use. Event Analysis also exposes **gaps and contradictions**, for instance a file shown as deleted on the live system but present in a later snapshot, or a modification time after the deletion, either of which signals an error or manipulation that must be explained. Logon events tell us which account was active during each cluster, and Windows Update events explain why a shadow copy appeared at an unusual time. Above all, it lets the examiner test competing explanations, such as "the history was cleared by an automatic setting" against "the history was cleared by hand", by looking at what surrounds the event.

Q4. Cloud Analysis and Lead Analysis

Cloud Analysis looks for signs that data was stored in, synchronised with or sent through cloud services. On the endpoint this means sync-client databases and logs (OneDrive, Google Drive, Dropbox), cached authentication tokens, browser history and cache for cloud web pages, and upload records. It helps answer whether the workbook was uploaded to a cloud service or sent through web mail, and if so when and from which account; which cloud accounts are configured on the device and whether any are personal; whether a sync client still holds copies or propagated a deletion; and what legal request would be needed to obtain the provider's records. Here, Cloud Analysis of the Edge history recovered from SC3 shows a personal web-mail compose page open from 21:15 to 21:18 on 10 Sep, with an attachment upload at 21:17. No sync client is installed.

Lead Analysis pulls out and links the identifiers that point investigators towards **new people, accounts, devices or places**: e-mail addresses, phone numbers, user names, URLs, IP addresses, device serial numbers, contacts and recurring keywords. It answers questions such as who else might be involved and to whom the file could have been sent, which other devices or storage media should be examined, whether there are links between the suspect and the competitor through shared contacts, domains or numbers, and which accounts need preservation requests before their data expires. In this case it surfaced three new leads: the personal web-mail address, a SanDisk Ultra Fit USB serial number that the company never issued, and a contact in the competitor's domain stored in Edge's autofill data.

Q5. How Volume Shadow Copy Analysis Recovers and Compares Earlier Versions

The Volume Shadow Copy Service takes point-in-time snapshots of an NTFS volume using **copy-on-write** [12]. After a snapshot is taken, any block on the live volume that is about to change is first copied, in its original state, into the shadow-copy store inside `System Volume Information`. A snapshot is reconstructed by reading changed blocks from that store and unchanged blocks from the live volume. Each snapshot therefore shows the volume exactly as it was at its creation time, including files later deleted, earlier versions of edited documents, and older copies of registry hives, event logs and browser databases.

Each property of VSCs maps onto a question in this case. Because SC3 (11 Sep, 13:05) was taken after the final edit but before the deletion, it still contains `Ennore_Tender_Rates.xlsx`, so the file can be **recovered**. SC2 holds an earlier draft with three worksheets while SC3 holds the final version with five, so **comparing versions** shows how the document developed. The Edge `History` database in SC3 still holds the browsing records that were **later cleared**. And SC4 (16 Sep) no longer contains the workbook and has a near-empty `History` file, which **proves the absence** and pins the deletion and clearance to the window between 11 Sep 13:05 and 16 Sep 02:00.

The recovery workflow used in this case runs in eight steps:

1. **Verify the image.** Re-check the E01 hash before any work.
2. **Locate the VSS store.** Find the catalog and store files in `System Volume Information`.
3. **Enumerate snapshots.** List SC1–SC4 with their store GUIDs and creation times, converted from UTC to IST.
4. **Expose each snapshot read-only.** Break each one down in ILookIX, in full or differential mode.
5. **Compare versions.** Diff each snapshot against the live volume, and SC2 against SC3.
6. **Extract the targets.** Export the workbook, the Edge `History` database and the relevant `$MFT` records from SC3.
7. **Hash and verify.** Compute SHA-256 and confirm the result with a second tool.
8. **Document.** Record store GUIDs, paths, hashes, tool versions and times.

On the tooling side, ILookIX can break VSCs down in full or differential mode, show their contents in its normal file-system browser, and recover deleted files and metadata from them [8]. To verify the result, the same snapshot is exposed with an independent tool, such as the open-source libvshadow `vshadowmount` on a read-only copy of the image, and the hashes of the extracted files are compared.

Q6. Hypothetical Timeline: Created, Modified, Copied, Deleted, Recovered

Date / time (IST)	Event	Evidence source
01 Sep 10:14	Workbook created in `C:\Users\revathi.l\Documents\Tenders\`	`$MFT` (`$FN` created) in SC2/SC3; Office `docProps/core.xml`
03 Sep 02:00	**SC2** snapshot captures the draft version (3 sheets)	VSC catalog
08 Sep 17:52	Final modification (5 sheets, final rates)	`$MFT` modified time; `core.xml` lastModifiedBy "Revathi L"
08 Sep 18:05	Final version e-mailed internally to the Commercial Manager	Company mail server (reference copy)
10 Sep 21:12	Copied to a personal USB drive (SanDisk Ultra Fit) as `E:\Ennore_Tender_Rates.xlsx`	LNK file with the USB volume serial; USBSTOR; Event 1006
10 Sep 21:17	Attachment uploaded on a personal web-mail compose page	Edge `History` recovered from SC3
11 Sep 13:05	**SC3** snapshot captures the final workbook and Edge history	VSC catalog; Windows Update log
11 Sep 18:44	Workbook deleted to the Recycle Bin (renamed `$RAB12CD.xlsx`)	`$UsnJrnl`: RENAME_NEW_NAME
11 Sep 18:46	Recycle Bin emptied	`$UsnJrnl`: FILE_DELETE for `$RAB12CD.xlsx` and `$IAB12CD.xlsx`
11 Sep 18:49	Edge "Clear browsing data – All time"	Edge `Preferences` timestamp; new empty `History`
16 Sep 02:00	**SC4** snapshot: workbook absent	VSC catalog

Date / time (IST)	Event	Evidence source
22 Sep 15:30	Workbook **recovered from SC3** and hashed	Examiner's notes; ILookIX export log

Q7. Validating the Recovered Artefact

A recovered file becomes a finding only once it has been shown to be **authentic, complete, correctly dated and reproducible**. The checklist below was applied to the workbook from SC3.

#	Validation check	Method	Result
1	Image integrity	Re-verify the E01 image hash before export	Match
2	Snapshot identity	Record the SC3 store GUID and creation time from the VSC catalog	Recorded
3	Hash of recovered file	SHA-256 of the exported workbook	`69ddd0b2…601fcd78`
4	Independent reference	Compare with the SHA-256 of the internal e-mail attachment (08 Sep) on the mail server	**Identical**
5	Second-tool verification	Expose SC3 with libvshadow on a copy; extract and hash the same file	**Identical**
6	File integrity	Open in a read-only viewer on an isolated machine; check the ZIP/XML structure	Opens cleanly, 5 sheets
7	Internal metadata	Check creator, lastModifiedBy, created and modified times in `docProps/core.xml`	Consistent with `$MFT`
8	File-system metadata	Compare `$SI` and `$FN` times in the SC3 `$MFT` record	No timestomping
9	Context	Confirm path and owner SID, and that the deletion appears in the live volume's `$UsnJrnl`	Consistent
10	Peer review	A second examiner repeats checks 3–5 from the notes	Same result
11	Documentation	Record tools, versions, commands and times so the result is reproducible	Complete

Q8. Sources of Error and Alternative Explanations

Several things could mislead an examiner reading these timestamps and deleted artefacts, and each was checked:

- **Snapshot time is not event time.** A file found in SC3 proves only that it existed at 11 Sep 13:05, not when it was created or deleted; those times must come from `$MFT`, `$UsnJrnl` and the logs.
- **Time zones and clocks.** VSC creation times, NTFS times, `$UsnJrnl`, Office metadata and LNK files are all in UTC, while Edge counts microseconds from 1601. A wrong display time zone shifts everything by 5 h 30 min.
- **Copy semantics.** Copying the file to the USB gives the *copy* a new created time while usually keeping the original modified time, so the copy can look as if it was "created after it was modified".
- **NTFS tunnelling.** If a file is deleted and a file with the same name is created within 15 seconds, Windows reuses the old creation time.
- **Unreliable last-access times.** These may be disabled, or updated by antivirus scans and indexing, so they cannot prove that a person opened a file.
- **Automatic clean-up.** Edge's "clear browsing data on close" option, Storage Sense or a scheduled cleaner could delete history or empty the Recycle Bin without any deliberate act. The `Preferences` file was checked: it records a manual "All time" clearance and no clear-on-close setting.
- **Other users or remote access.** Someone else with her password, or a remote-management session, could have performed the actions, so logon events, remote-tool logs and CCTV were reviewed.
- **Office temporary and AutoRecover files.** Files such as `~$Ennore_Tender_Rates.xlsx` generate activity that is not user action.
- **Incomplete snapshots.** Copy-on-write keeps only changed blocks, so a recovered file could be partial if a block was never preserved; the hash match with an independent copy rules this out here.

- **Tool interpretation.** Different tools can parse VSCs or browser timestamps differently, which is why two-tool verification was used.

Conclusion

The workbook `Ennore_Tender_Rates.xlsx` was recovered from Volume Shadow Copy SC3 and is byte-for-byte identical to the final version e-mailed internally on 08 Sep. The artefacts show that it was copied to a personal USB drive and attached to a personal web-mail message on 10 Sep, and that on 11 Sep, within five minutes, the file was deleted, the Recycle Bin emptied and the browser history cleared by hand. These findings are consistent with deliberate removal of evidence by the `revathi.l` account. Before any conclusion is drawn about the individual, I would recommend examining the SanDisk USB drive, sending a preservation request to the web-mail provider, and conducting a formal interview.

Reflection

What surprised me most here was that the strongest evidence came from a feature nobody in the case was thinking about: a shadow copy created by a routine Windows update a few hours before the deletion. It showed me why an examiner should always check what the operating system keeps on its own, and why a recovered file needs an independent reference, here the internal e-mail, before it is trusted.

ASSIGNMENT – 2(5)

Multi-Source Investigation of a Suspected Cyber Incident

Unique Scenario: "The 1 AM Leaderboard". Unauthorised score changes on the cloud-hosted judging dashboard of the inter-college hackathon "CodeSprint 2026", made through a student volunteer's account. The event, platform and people are fictional.

Case No. CS26/DFC/2026/046 · Incident window: Sat 03 Oct 2026, 01:08–01:31 IST · Examiner: M. M. Chandhan Sai

Case Scenario

CodeSprint 2026 used a cloud SaaS platform, **HackHub** (fictional), for submissions and judging. On the morning of 3 October the organisers noticed that **Team-07 "ByteBrigade"** had climbed from sixth place to second overnight, after its scores rose in three categories. HackHub's audit log showed the changes were made between **01:13 and 01:26 IST** by the account `vol-0419`, which belongs to a third-year student volunteer, **Ms. K. Harini**. Volunteers are only meant to view the dashboard, but an organiser had mistakenly given the volunteer role write permission on scores while making a "quick fix" on 1 October.

At 07:15 Harini e-mailed the organisers herself. She had received a "new sign-in" alert, she said, but had been asleep and had not logged in. With her written consent and the college's authorisation, investigators obtained her **Windows laptop**, her **Chrome browser profile**, her **Gmail** messages, her **WhatsApp** records, her **Android phone**, and **cloud and network records** from HackHub and the hostel network.

Why I chose this scenario. Most incidents students are actually involved in today span a laptop, a phone and several online services, and no single device tells the whole story. A hackathon leaderboard is familiar to any CS student, and the student's own report that she was asleep gives the investigation a real hypothesis to test rather than a foregone conclusion.

Source	Device / record
Windows computer	HP Pavilion 14 laptop, Windows 11 Home 23H2, Windows Hello PIN sign-in
Browser	Google Chrome 129, profile "Harini" (passwords saved and synced)
E-mail	Gmail account, used through the browser and the Android app
Messaging	WhatsApp on the phone (end-to-end encrypted)
Mobile phone	Samsung Galaxy M34, Android 14
Cloud	HackHub audit log, admin change log and role-permission history (UTC)
Network	Hostel firewall NAT log and DHCP server log (IST)

Assumptions

- All people, accounts, IP addresses, platform names and values are fictional. Harini consented in writing for her devices and accounts; the organisers preserved HackHub's logs on 3 Oct and exported them with hashes.
- Team-07 member Ms. D. Swetha is Harini's roommate. Her devices are **not** available at this stage.
- Tool operations describe the intended workflow.

Q1. Evidence from Windows and Other Operating Systems

Windows laptop. The Security event log shows Event 4801 (workstation unlocked) and Event 4624 with Logon Type 7 (unlock) at 01:08:22, and the Windows Hello operational log confirms that a **PIN** was used, not a fingerprint or face. In the System log, Kernel-Power Event 42 records the laptop going to sleep at 01:31:12, and WLAN-AutoConfig events show its connection to the hostel Wi-Fi. The registry adds network profiles (SSID, first and last connection), USB history and recently used files, while the SRUM database (`SRUDB.dat`) records per-application network use by the hour and shows Chrome moving data between 01:00 and 02:00. Prefetch, Jump Lists and LNK files confirm that Chrome was launched and which files were opened. Beyond these sit the Chrome profile itself (Q2), Windows Timeline (`ActivitiesCache`) if it is enabled, and memory remnants in `pagefile.sys` and `hiberfil.sys`.

Other operating systems. The **Android 14** phone holds usage statistics (screen on and off, which app was in the foreground), Digital Wellbeing data, battery and charging history, notification history, SMS, the WhatsApp database, the Gmail app cache, Google account sync records, Wi-Fi history and location history. On the server side, HackHub's

web and API logs and the hostel firewall, a Linux-based appliance, provide NAT and DHCP logs. These are collected by the providers and handed over with integrity hashes.

Q2. What Browser Artefacts Reveal, and Why They Need Careful Interpretation

Artefact (Chrome profile)	What it reveals in this case
`History`: urls/visits tables	Visits to `judge.hackhub.example/login` at 01:10:55, `/dashboard/scores/team-07` between 01:13 and 01:26, and `/leaderboard` at 01:26:30; the first visit was "typed"
`Login Data`	Saved credential for `vol-0419`; `times_used` incremented and `date_last_used` 01:11:03, so autofill was used
`Cookies`	HackHub session cookie created at 01:11:07, matching the cloud login time
`Web Data` (autofill)	Form entries; no new score values typed into stored forms
Cache	Cached copies of the score-edit page and API responses
`Preferences` / Sync data	Profile signed in to Harini's Google account, with password sync on
Sessions / Tabs	Tab layout restored at the next start, showing what was open

Browser evidence is detailed, but it has to be read with care. It shows what the **profile** did, not who was sitting at the keyboard: anyone using the unlocked laptop uses Harini's profile and her saved passwords, and with autofill nobody even needs to know the password to log in. Because Chrome syncs across devices, a history entry may have been created on another device signed into the same Google account; Chrome's sync metadata records which device, and here every visit is local. Pre-fetching and background requests can create entries the user never clicked, so typed and linked visits must be told apart. Chrome stores time as microseconds since 1 January 1601 (UTC), and a careless conversion gives wrong times. Incognito browsing, cleared history and multiple profiles can all leave gaps, so an absence of history does not prove an absence of activity. Finally, a cache timestamp records when content was stored, which is not necessarily when it was viewed.

Q3. Evidence from E-mail and Messaging, and How It Correlates

E-mail (Gmail) can provide message content and attachments, and above all the **full headers**: the `Received:` chain, `Date`, `Message-ID`, sending IP and DKIM/SPF results. Service notifications are especially useful here. HackHub's alert "New sign-in from Chrome on Windows" arrived at **01:11:09**, and its read status shows it was opened at **07:02** in the Android Gmail app. Google account security events (sign-ins, devices, recovery changes) can be obtained by consent from the account's security page or a Takeout export, and the headers of Harini's own 07:15 report show it was sent from her phone.

Messaging (WhatsApp) stores messages, timestamps, the numbers of senders and recipients, group membership, delivery and read receipts, media, call logs and placeholders for deleted messages, all in the phone's `msgstore.db` database. Two messages matter most: at **02 Oct 23:40** Swetha wrote, *"can I use your laptop tonight to finish our slides? my charger died"*, and at **23:44** Harini replied, *"ok, PIN same as before"*. Since WhatsApp content is end-to-end encrypted, the provider holds very little, and the phone is the main source.

These sources become far more valuable once they are **correlated**. The HackHub alert at 01:11:09 matches both the HackHub login (19:41:07 UTC, which is 01:11:07 IST) and the creation of the Chrome cookie at 01:11:07. The e-mail's read time of 07:02 matches the phone's first screen-on after charging. And the WhatsApp exchange explains who had permission to use the laptop and knew its PIN, which connects directly to the PIN unlock at 01:08:22. With every e-mail and message time converted to IST, all of this falls into a single timeline (the correlation matrix in Q6).

Q4. Forensic Considerations and Challenges for the Mobile Phone

A phone holds far more private data than any incident needs, including other people's messages, so **authority and scope** come first. It is examined only with consent or legal authority, and only for the relevant period and data types, here 02 Oct 20:00 to 03 Oct 08:00 [4]. The phone is then **isolated**, either in airplane mode or in a Faraday bag, so that new messages, a remote wipe or cloud sync cannot change its contents, and it is kept charged so it does not power off and lock again. The **lock state** is preserved carefully: Android 14 uses file-based encryption, under which most data is available only after the first unlock, so the examiner records whether the phone is "before first unlock" or "after first unlock" and does not reboot it.

The **acquisition method** is chosen by need and tool support, from a logical extraction of app data through backup interfaces, to a file-system or full file-system extraction with a validated mobile forensic tool. Physical chip-off is rarely useful on modern encrypted phones. The IMEI, model, OS version, security patch level, SIM or eSIM details, battery state and the tool and version used are all recorded, and the extraction output is hashed. In this case, the key artefacts are the usage statistics showing the **screen off and charging from 00:31 to 06:42**, the WhatsApp messages, and the Gmail notification and its read time. The SMS store holds no HackHub one-time password, because HackHub did not use two-factor authentication.

Mobile evidence brings its own **challenges**. Strong device encryption and secure hardware can make extraction impossible without the passcode. Frequent OS and app updates break tool support and change database formats, and the sheer variety of devices, chipsets and apps makes tool coverage uneven. Encrypted messaging, disappearing messages and encrypted local backups (WhatsApp's `.crypt15` files) limit what can be read. Data keeps changing as the phone syncs with the cloud, and notifications and logs roll over quickly. Remote-wipe and anti-forensic apps are a risk, and the privacy of third parties whose messages sit on the phone must be protected throughout.

Q5. Internet and Cloud-Based Evidence Sources

Source	Relevant content	How obtained
HackHub audit log	`LOGIN_SUCCESS vol-0419` at 19:41:07Z from IP 117.193.44.12, User-Agent Chrome/129 on Windows; `SCORE_UPDATE` team-07 at 19:43:40Z, 19:49:02Z and 19:55:51Z	Exported by the organiser admin; hashed on 03 Oct; provider certificate
HackHub admin change log	Volunteer role given `scores:write` on 01 Oct at 18:20 IST by an organiser account	As above
Hostel firewall NAT log	10.20.14.88:51544 → HackHub IP at 01:11:05 IST, translated to public IP 117.193.44.12	Exported by the college network team, hashed
Hostel DHCP log	10.20.14.88 leased to MAC 3C:91:80:6B:2E:14 (Harini's laptop Wi-Fi) since 22:10	As above
Google account	Sign-in and device activity, Gmail headers, Chrome sync metadata	Consent-based export; legal request if needed
Meta / WhatsApp	Limited subscriber and connection metadata	Legal process only; content not held server-side
Cloud backups	WhatsApp backup on Google Drive; Google Photos	Consent; usually unnecessary when the device is available
DNS / proxy logs	Name lookups for `judge.hackhub.example`	College network team

Cloud evidence must be **preserved early**, because providers keep logs only for limited periods. The organisers' export on 3 Oct is the reason the HackHub audit trail survived at all.

Q6. Cross-Source Correlation Matrix

#	Source	Artefact	Time (IST)	Identifier	Correlates with
C1	Mobile	WhatsApp: Swetha asks to use the laptop; Harini agrees, "PIN same as before"	02 Oct 23:40 / 23:44	Two phone numbers	C3 (PIN unlock); explains opportunity
C2	Mobile	Usage stats: screen off, charging	00:31 – 06:42	Device IMEI	C3–C7 all occur while the phone is idle
C3	Computer	Event 4801 / 4624 Type 7: PIN unlock	01:08:22	Laptop, user `harini`	C4 (Chrome starts two minutes later)
C4	Browser	Chrome visits `/login`; saved password autofilled	01:10:55 / 01:11:03	Chrome profile "Harini"	C5 (login), C6 (NAT)
C5	Cloud	HackHub `LOGIN_SUCCESS vol-0419`	19:41:07Z = 01:11:07	IP 117.193.44.12	C6 (NAT mapping), C7 (alert e-mail)

#	Source	Artefact	Time (IST)	Identifier	Correlates with
C6	Network	NAT 10.20.14.88 → 117.193.44.12; DHCP lease to laptop MAC	01:11:05	MAC 3C:91:80:6B:2E:14	Ties C5 to Harini's laptop
C7	E-mail	HackHub "New sign-in" alert received; read at 07:02 on phone	01:11:09 / 07:02	Gmail Message-ID	C5; C2 (phone idle until 06:42)
C8	Cloud	`SCORE_UPDATE` team-07 (×3)	01:13:40, 01:19:02, 01:25:51	Session of vol-0419	C9 (browser visits the same pages)
C9	Browser	Visits to `/dashboard/scores/team-07` and `/leaderboard`	01:13 – 01:26:30	Chrome profile	C8
C10	Computer	Kernel-Power 42: laptop sleeps	01:31:12	Laptop	End of the activity cluster

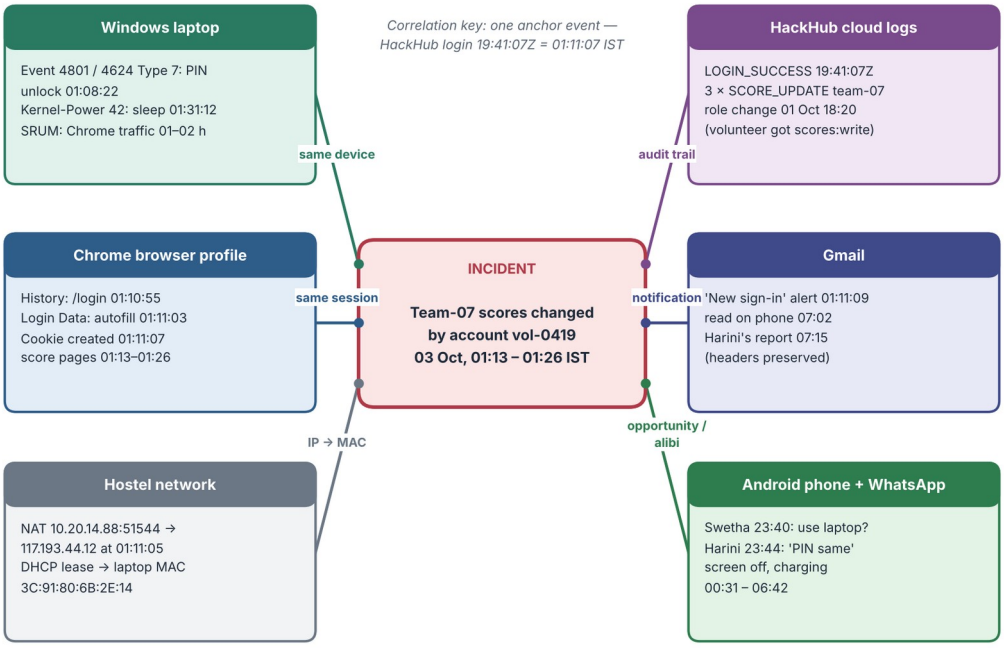


Figure 5.1: Multi-source evidence map linking every source to the incident

Q7. Challenges When Evidence Is Spread Across Devices and Cloud Services

Challenge	How it appears in this case	Mitigation
1. Attribution	The account, laptop and browser profile all belong to Harini, yet someone else may have used them	Look for independent evidence of the person: phone activity, messages, interviews, physical access
2. Time synchronisation and time zones	HackHub logs in UTC, firewall in IST, Chrome in 1601-epoch UTC, phone in local time	Normalise to IST in one documented step; check against an anchor event (login, NAT entry, cookie)
3. Encryption	WhatsApp end-to-end encryption; Android file-based encryption; BitLocker on many laptops	Consent-based unlock; device-level extraction; record what could not be accessed
4. Jurisdiction and legal process	Cloud providers may be outside India and require formal requests	Early preservation requests; providers' law-enforcement processes; consent where possible
5. Data volume and variety	Dozens of artefact types across five sources	Scope by time window and identifiers; timeline tools; filter known data

Challenge	How it appears in this case	Mitigation
6. Volatility and retention	Cloud audit logs and firewall logs are kept only for days or weeks	Preserve on day one, with hashes
7. Shared and NATed IP addresses	The hostel NAT puts hundreds of students behind one public IP	Use NAT and DHCP logs to map public IP → private IP → MAC → device
8. Chain of custody across organisations	Logs were exported by HackHub's organisers and the college network team	Obtain export details, hashes and certificates from each provider
9. Privacy and proportionality	The phone and chats contain friends' and family members' data	Limit extraction to the relevant time and apps; redact third-party data
10. Tool reliability	Different tools parse Chrome and Android data differently	Validate with a second tool; document versions [13]

Q8. Final Investigation Conclusion

Direct evidence. HackHub's audit log records that `vol-0419` logged in at 01:11:07 IST and changed Team-07's scores three times between 01:13 and 01:26. The network logs map that login's public IP and port to the private address 10.20.14.88 and then to the **MAC address of Harini's laptop**, so the actions were carried out **from her laptop**. Her Chrome profile contains visits to the same score-edit pages and use of the saved `vol-0419` password at the same times.

Corroborating evidence. The laptop was unlocked with the Windows Hello **PIN** at 01:08:22, two minutes before the login, and went to sleep at 01:31. The HackHub sign-in alert at 01:11:09 matches the login and went unread until 07:02. Harini's phone was idle and charging from 00:31 to 06:42, which fits her account of being asleep. The WhatsApp messages show that she allowed her roommate Swetha, a Team-07 member, to use the laptop that night, and that Swetha knew the PIN. Finally, the organiser's role change on 1 October explains how a volunteer account could edit scores in the first place.

Unresolved questions. The central question is still open: **who physically operated the laptop between 01:08 and 01:31?** The digital evidence identifies the laptop and the account, not the person. Swetha's devices and account activity have not been examined, and doing so requires her consent or proper authority. There is no CCTV or witness evidence for the hostel room, and it is not yet known whether other volunteer accounts with write access were also misused.

Overall conclusion. The score changes were made through Harini's account, from her laptop, using her browser's saved password. However, her idle phone, together with messages showing that another person had both permission and the PIN, means the investigation **cannot attribute the actions to Harini personally**. I would recommend restoring the original scores from HackHub's audit trail, removing write permission from the volunteer role, enabling two-factor authentication for judging accounts, and referring the question of who used the laptop to the organising committee, where Swetha can be heard.

Reflection

This was the case where correlation mattered most. Each source on its own looked damning or reassuring depending on where you started, but anchoring everything to a single event, the HackHub login at 01:11:07, turned six separate record sets into one consistent story. It also showed me how easily a person's honest report ("I was asleep") could be dismissed if an investigator stopped at the account logs.

REFERENCES

Standards, Laws and Literature Cited

Numbers in square brackets in the text refer to the sources below. The descriptions of ILookIX and IXImager are based on [6]–[8] together with the CR23532 unit content.

[1] K. Kent, S. Chevalier, T. Grance and H. Dang, *Guide to Integrating Forensic Techniques into Incident Response*, NIST Special Publication 800-86, National Institute of Standards and Technology, 2006.

[2] D. Brezinski and T. Killalea, *Guidelines for Evidence Collection and Archiving*, RFC 3227, Internet Engineering Task Force, 2002.

[3] Association of Chief Police Officers (ACPO), *Good Practice Guide for Digital Evidence*, Version 5, 2012.

[4] ISO/IEC 27037:2012, *Guidelines for Identification, Collection, Acquisition and Preservation of Digital Evidence*, International Organization for Standardization, 2012.

[5] B. Carrier, *File System Forensic Analysis*, Addison-Wesley, 2005.

[6] National Institute of Justice, *Test Results for Digital Data Acquisition Tool: IXImager (Version 2.0, Feb-01 2006)*, NCJ 217678, U.S. Department of Justice, 2007.

[7] S. Garfinkel, D. Malan, K.-A. Dubec, C. Stevens and C. Pham, "Advanced Forensic Format: An Open, Extensible Format for Disk Imaging," in *Advances in Digital Forensics II* (IFIP WG 11.9), Springer, 2006.

[8] Forensics Wiki, entries "ILook", "IXimager" and "IXimager file formats," forensics.wiki, accessed October 2026.

[9] Government of India, *The Information Technology Act, 2000* (Act No. 21 of 2000), Sections 43 and 66.

[10] Government of India, *The Bharatiya Sakshya Adhiniyam, 2023* (Act No. 47 of 2023), Section 63.

[11] Government of India, *The Digital Personal Data Protection Act, 2023* (Act No. 22 of 2023).

[12] Microsoft, "Volume Shadow Copy Service," Microsoft Learn documentation, accessed October 2026.

[13] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers and the Internet*, 3rd ed., Academic Press, 2011.